

a09_transformers_mlflow_answers

August 15, 2026

1 Assignment 09: Transformers and MLflow: Downstream Tasks and Experiment Tracking

Name	Vishani Raveendran
IIT Number	20260276
Course	AI Tools and Concepts

1.1 Setup

Load the shared libraries and point MLflow at a local SQLite tracking store (`./mlflow.db`) inside this assignment folder, so `mlflow ui` can be launched from here to inspect every run.

```
import os
import pandas as pd
import mlflow
from transformers import pipeline, set_seed

mlflow.set_tracking_uri("sqlite:/// " + os.path.abspath("mlflow.db"))
print("MLflow tracking URI:", mlflow.get_tracking_uri())
```

MLflow tracking URI:

sqlite:///Users/vishaniraveendran/Library/CloudStorage/GoogleDrive-vishani.20260276@iit.ac.lk/My Drive/MS-C-Vish/First Year/Ai Tools and Concepts/Assignment/Assignments/A09/mlflow.db

1.2 Task 1: Sentiment Classification

Use the pretrained `distilbert-base-uncased-finetuned-sst-2-english` model (via the Hugging Face `pipeline` API) to classify custom sentences as `POSITIVE` / `NEGATIVE`. Each batch of sentences is self-labeled with an expected sentiment so accuracy can be computed, then every run is tracked in MLflow: model name, sample count, accuracy, and the predictions CSV as an artifact. Two runs are logged — a straightforward batch and a harder batch with negation/sarcasm — to compare how accuracy shifts with sentence difficulty.

```
sentiment_pipe = pipeline(
    "sentiment-analysis",
```

```

    model="distilbert-base-uncased-finetuned-sst-2-english",
    device=-1,
)

def run_sentiment_experiment(run_name, sentences, true_labels):
    predictions = sentiment_pipe(sentences)
    pred_labels = [p["label"] for p in predictions]
    pred_scores = [p["score"] for p in predictions]
    accuracy = sum(p == t for p, t in zip(pred_labels, true_labels)) / len(true_labels)

    df = pd.DataFrame({
        "sentence": sentences,
        "true_label": true_labels,
        "predicted_label": pred_labels,
        "confidence": pred_scores,
    })

    artifact_path = f"{run_name}_predictions.csv"
    df.to_csv(artifact_path, index=False)

    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", "distilbert-base-uncased-finetuned-sst-2-english")
        mlflow.log_param("num_samples", len(sentences))
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_artifact(artifact_path)

    os.remove(artifact_path)
    print(f"[{run_name}] accuracy = {accuracy:.2%}")
    return df

mlflow.set_experiment("A09_Task1_Sentiment_Classification")

```

Loading weights: 100% | 104/104 [00:00<00:00, 7223.65it/s]

<Experiment:

```

artifact_location=('/Users/vishaniraveendran/Library/CloudStorage/GoogleDrive-vishani.20260276@iit.ac.lk/My '
'Drive/MS-CVish/First Year/Ai Tools and '
'Concepts/Assignment/Assignments/A09/mlruns/1'), creation_time=1785590050487,
effective_trace_archival_retention=None, experiment_id='1',
last_update_time=1785590050487, lifecycle_stage='active',
name='A09_Task1_Sentiment_Classification', tags={}, trace_location=None,
workspace='default'>

```

```

# Run 1: clear-cut sentences (10 positive, 10 negative)
sentences_run1 = [
    "The movie was absolutely fantastic and kept me on the edge of my seat.",
    "I loved the customer service at this restaurant.",
    "This is the best phone I have ever owned.",
    "The concert last night was an unforgettable experience.",
    "Her presentation was clear, engaging, and well organized.",
    "I am so happy with how my new apartment turned out.",
    "The team did an amazing job finishing the project early.",
    "What a wonderful sunny day for a picnic.",
    "The book was a delightful and heartwarming read.",
    "Our vacation in Bali exceeded all of our expectations.",
    "The food was cold and the service was incredibly slow.",
    "I regret buying this laptop, it crashes constantly.",
    "The traffic this morning was a nightmare.",
    "This is the worst customer support I have ever dealt with.",
    "The hotel room smelled awful and was not cleaned properly.",
    "I was extremely disappointed with the ending of the film.",
    "The product broke after just two days of use.",
    "Waiting in line for three hours was frustrating and exhausting.",
    "The lecture was boring and difficult to follow.",
    "I can't believe how rude the staff were to us.",
]
labels_run1 = ["POSITIVE"] * 10 + ["NEGATIVE"] * 10

df_run1 = run_sentiment_experiment("distilbert_run1_general", sentences_run1,
    labels_run1)
df_run1

```

```
[distilbert_run1_general] accuracy = 100.00%
```

	sentence	true_label	\
0	The movie was absolutely fantastic and kept me...	POSITIVE	
1	I loved the customer service at this restaurant.	POSITIVE	
2	This is the best phone I have ever owned.	POSITIVE	
3	The concert last night was an unforgettable ex...	POSITIVE	
4	Her presentation was clear, engaging, and well...	POSITIVE	
5	I am so happy with how my new apartment turned...	POSITIVE	
6	The team did an amazing job finishing the proj...	POSITIVE	
7	What a wonderful sunny day for a picnic.	POSITIVE	
8	The book was a delightful and heartwarming read.	POSITIVE	
9	Our vacation in Bali exceeded all of our expec...	POSITIVE	
10	The food was cold and the service was incredib...	NEGATIVE	
11	I regret buying this laptop, it crashes consta...	NEGATIVE	
12	The traffic this morning was a nightmare.	NEGATIVE	
13	This is the worst customer support I have ever...	NEGATIVE	
14	The hotel room smelled awful and was not clean...	NEGATIVE	

15	I was extremely disappointed with the ending o...	NEGATIVE
16	The product broke after just two days of use.	NEGATIVE
17	Waiting in line for three hours was frustratin...	NEGATIVE
18	The lecture was boring and difficult to follow.	NEGATIVE
19	I can't believe how rude the staff were to us.	NEGATIVE

	predicted_label	confidence
0	POSITIVE	0.999883
1	POSITIVE	0.999818
2	POSITIVE	0.999728
3	POSITIVE	0.999820
4	POSITIVE	0.999872
5	POSITIVE	0.999876
6	POSITIVE	0.999601
7	POSITIVE	0.999887
8	POSITIVE	0.999887
9	POSITIVE	0.998372
10	NEGATIVE	0.999796
11	NEGATIVE	0.998787
12	NEGATIVE	0.999541
13	NEGATIVE	0.999777
14	NEGATIVE	0.999800
15	NEGATIVE	0.999789
16	NEGATIVE	0.999450
17	NEGATIVE	0.999413
18	NEGATIVE	0.999730
19	NEGATIVE	0.999518

```
# Run 2: harder sentences with negation, mixed sentiment, and sarcasm (10
↳positive, 10 negative)
sentences_run2 = [
    "The plot wasn't bad, but it definitely could've been better.",
    "I don't dislike the new update, it's actually pretty solid.",
    "Oh great, another Monday morning meeting that could've been an email.",
    "Sure, because waiting two hours for a table is exactly what I wanted.",
    "Not the worst meal I've had, but far from the best either.",
    "I wouldn't say I hated it, but I probably won't watch it again.",
    "The service wasn't great, however the food made up for it.",
    "It's hard to complain about a hotel this comfortable.",
    "I was skeptical at first, but the movie completely won me over.",
    "The instructions were confusing, yet somehow the product still works well.
↳",
    "Yeah, losing my luggage on day one was a fantastic start to the trip.",
    "This is far from perfect, but it gets the job done.",
    "I can't say I'm impressed, but it wasn't a total disaster.",
    "The staff couldn't have been more unhelpful if they tried.",
    "It's not that the coffee was bad, it's that it was overpriced.",
```

```

    "Honestly, I expected far worse given the reviews.",
    "The app rarely crashes anymore, which is a nice change.",
    "I guess it's fine if you enjoy paying extra for nothing.",
    "Despite the rocky start, the whole experience turned out great.",
    "The design looks nice, but the battery life ruins the experience.",
]
labels_run2 = [
    "NEGATIVE", "POSITIVE", "NEGATIVE", "NEGATIVE", "NEGATIVE",
    "NEGATIVE", "POSITIVE", "POSITIVE", "POSITIVE", "POSITIVE",
    "NEGATIVE", "POSITIVE", "NEGATIVE", "NEGATIVE", "NEGATIVE",
    "POSITIVE", "POSITIVE", "NEGATIVE", "POSITIVE", "NEGATIVE",
]

df_run2 = run_sentiment_experiment("distilbert_run2_nuanced", sentences_run2,
    ↪labels_run2)
df_run2

```

```
[distilbert_run2_nuanced] accuracy = 75.00%
```

	sentence	true_label	\
0	The plot wasn't bad, but it definitely could've...	NEGATIVE	
1	I don't dislike the new update, it's actually ...	POSITIVE	
2	Oh great, another Monday morning meeting that ...	NEGATIVE	
3	Sure, because waiting two hours for a table is...	NEGATIVE	
4	Not the worst meal I've had, but far from the ...	NEGATIVE	
5	I wouldn't say I hated it, but I probably won'...	NEGATIVE	
6	The service wasn't great, however the food mad...	POSITIVE	
7	It's hard to complain about a hotel this comfo...	POSITIVE	
8	I was skeptical at first, but the movie comple...	POSITIVE	
9	The instructions were confusing, yet somehow t...	POSITIVE	
10	Yeah, losing my luggage on day one was a fanta...	NEGATIVE	
11	This is far from perfect, but it gets the job ...	POSITIVE	
12	I can't say I'm impressed, but it wasn't a tot...	NEGATIVE	
13	The staff couldn't have been more unhelpful if...	NEGATIVE	
14	It's not that the coffee was bad, it's that it...	NEGATIVE	
15	Honestly, I expected far worse given the reviews.	POSITIVE	
16	The app rarely crashes anymore, which is a nic...	POSITIVE	
17	I guess it's fine if you enjoy paying extra fo...	NEGATIVE	
18	Despite the rocky start, the whole experience ...	POSITIVE	
19	The design looks nice, but the battery life ru...	NEGATIVE	

	predicted_label	confidence
0	NEGATIVE	0.995218
1	POSITIVE	0.999856
2	NEGATIVE	0.988933
3	POSITIVE	0.983871
4	NEGATIVE	0.981867

5	NEGATIVE	0.992762
6	POSITIVE	0.997563
7	POSITIVE	0.999648
8	POSITIVE	0.999381
9	POSITIVE	0.999620
10	POSITIVE	0.998887
11	POSITIVE	0.995213
12	POSITIVE	0.990352
13	NEGATIVE	0.981555
14	NEGATIVE	0.997980
15	NEGATIVE	0.998030
16	POSITIVE	0.998240
17	POSITIVE	0.997861
18	POSITIVE	0.999766
19	NEGATIVE	0.999705

Observations: Run 1 (clear-cut sentences) scored **100% accuracy** — DistilBERT is very confident (>0.998) on unambiguous positive/negative language. Run 2 (negation, mixed sentiment, sarcasm) dropped to **75% accuracy** (15/20 correct), with 5 misclassifications concentrated in sentences that mix a positive and negative clause (e.g. “Sure, because waiting two hours for a table is exactly what I wanted”, “Honestly, I expected far worse given the reviews”). The model appears to key off surface-level positive words/phrases and struggles with sarcasm and double-negation, even though its confidence scores stay high (>0.98) even when wrong — showing it is confidently incorrect rather than uncertain on hard examples.

1.3 Task 2: Named Entity Recognition

Use the pretrained `dslim/bert-base-NER` model to extract **PER** (person), **ORG** (organization), and **LOC** (location) entities from paragraphs. Two runs of 10 paragraphs each are logged to MLflow — one news-style batch, one biography-style batch — recording the model name, total entity count, the distribution of entity types, and the extracted-entity table as a CSV artifact.

```
ner_pipe = pipeline(
    "ner",
    model="dslim/bert-base-NER",
    aggregation_strategy="simple",
    device=-1,
)

def run_ner_experiment(run_name, paragraphs):
    rows = []
    for i, para in enumerate(paragraphs):
        for ent in ner_pipe(para):
            rows.append({
                "paragraph_id": i,
                "paragraph": para,
                "entity": ent["word"],
```

```

        "label": ent["entity_group"],
        "score": float(ent["score"]),
    })
df = pd.DataFrame(rows)
distribution = df["label"].value_counts().to_dict()

artifact_path = f"{run_name}_entities.csv"
df.to_csv(artifact_path, index=False)

with mlflow.start_run(run_name=run_name):
    mlflow.log_param("model_name", "dslim/bert-base-NER")
    mlflow.log_param("num_paragraphs", len(paragraphs))
    mlflow.log_metric("total_entities", len(df))
    for label, count in distribution.items():
        mlflow.log_metric(f"count_{label}", count)
    mlflow.log_artifact(artifact_path)

os.remove(artifact_path)
print(f"[{run_name}] total entities = {len(df)} | distribution = {distribution}")
return df

mlflow.set_experiment("A09_Task2_Named_Entity_Recognition")

```

Loading weights: 100% | 199/199 [00:00<00:00, 5597.77it/s]
[transformers] BertForTokenClassification LOAD REPORT from: dslim/bert-base-NER

Key	Status		
bert.pooler.dense.bias	UNEXPECTED		
bert.pooler.dense.weight	UNEXPECTED		

Notes:

- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.

<Experiment:

```

artifact_location=('/Users/vishaniraveendran/Library/CloudStorage/GoogleDrive-vishani.20260276@iit.ac.lk/My '
'Drive/MS-C-Vish/First Year/Ai Tools and '
'Concepts/Assignment/Assignments/A09/mlruns/2'), creation_time=1785590105281,
effective_trace_archival_retention=None, experiment_id='2',
last_update_time=1785590105281, lifecycle_stage='active',
name='A09_Task2_Named_Entity_Recognition', tags={}, trace_location=None,
workspace='default'>

```

```
# Run 1: news-style paragraphs
paragraphs_run1 = [
    "Tim Cook, the CEO of Apple, unveiled the new iPhone at a press event in_
    ↪Cupertino, California.",
    "Serena Williams announced her retirement from professional tennis during_
    ↪an event hosted by ESPN in New York.",
    "The United Nations held an emergency session in Geneva to discuss the_
    ↪ongoing crisis in Ukraine.",
    "Elon Musk confirmed that Tesla will build a new Gigafactory near Berlin,_
    ↪Germany.",
    "Jeff Bezos stepped down as CEO of Amazon and was succeeded by Andy Jassy_
    ↪in Seattle.",
    "The World Health Organization released new guidelines from its_
    ↪headquarters in Geneva, Switzerland.",
    "Cristiano Ronaldo signed a new contract with Al Nassr, a football club_
    ↪based in Riyadh, Saudi Arabia.",
    "Google opened a new research lab in Bangalore, India, led by Sundar Pichai.
    ↪",
    "The European Central Bank raised interest rates during a meeting in_
    ↪Frankfurt, Germany.",
    "Taylor Swift performed a sold-out concert at Wembley Stadium in London.",
]

df_ner_run1 = run_ner_experiment("bert_ner_run1_news", paragraphs_run1)
df_ner_run1
```

```
[bert_ner_run1_news] total entities = 42 | distribution = {'LOC': 19, 'PER': 11,
'ORG': 11, 'MISC': 1}
```

	paragraph_id	paragraph \
0	0	Tim Cook, the CEO of Apple, unveiled the new i...
1	0	Tim Cook, the CEO of Apple, unveiled the new i...
2	0	Tim Cook, the CEO of Apple, unveiled the new i...
3	0	Tim Cook, the CEO of Apple, unveiled the new i...
4	0	Tim Cook, the CEO of Apple, unveiled the new i...
5	1	Serena Williams announced her retirement from ...
6	1	Serena Williams announced her retirement from ...
7	1	Serena Williams announced her retirement from ...
8	2	The United Nations held an emergency session i...
9	2	The United Nations held an emergency session i...
10	2	The United Nations held an emergency session i...
11	3	Elon Musk confirmed that Tesla will build a ne...
12	3	Elon Musk confirmed that Tesla will build a ne...
13	3	Elon Musk confirmed that Tesla will build a ne...
14	3	Elon Musk confirmed that Tesla will build a ne...
15	3	Elon Musk confirmed that Tesla will build a ne...

16	3	Elon Musk confirmed that Tesla will build a ne...
17	3	Elon Musk confirmed that Tesla will build a ne...
18	4	Jeff Bezos stepped down as CEO of Amazon and w...
19	4	Jeff Bezos stepped down as CEO of Amazon and w...
20	4	Jeff Bezos stepped down as CEO of Amazon and w...
21	4	Jeff Bezos stepped down as CEO of Amazon and w...
22	5	The World Health Organization released new gui...
23	5	The World Health Organization released new gui...
24	5	The World Health Organization released new gui...
25	6	Cristiano Ronaldo signed a new contract with A...
26	6	Cristiano Ronaldo signed a new contract with A...
27	6	Cristiano Ronaldo signed a new contract with A...
28	6	Cristiano Ronaldo signed a new contract with A...
29	6	Cristiano Ronaldo signed a new contract with A...
30	6	Cristiano Ronaldo signed a new contract with A...
31	7	Google opened a new research lab in Bangalore,...
32	7	Google opened a new research lab in Bangalore,...
33	7	Google opened a new research lab in Bangalore,...
34	7	Google opened a new research lab in Bangalore,...
35	7	Google opened a new research lab in Bangalore,...
36	8	The European Central Bank raised interest rate...
37	8	The European Central Bank raised interest rate...
38	8	The European Central Bank raised interest rate...
39	9	Taylor Swift performed a sold-out concert at W...
40	9	Taylor Swift performed a sold-out concert at W...
41	9	Taylor Swift performed a sold-out concert at W...

	entity	label	score
0	Tim Cook	PER	0.999788
1	Apple	ORG	0.998776
2	iPhone	MISC	0.994731
3	Cupertino	LOC	0.997840
4	California	LOC	0.999380
5	Serena Williams	PER	0.999558
6	ESPN	ORG	0.998583
7	New York	LOC	0.999358
8	United Nations	ORG	0.999354
9	Geneva	LOC	0.999682
10	Ukraine	LOC	0.999834
11	Elon Musk	ORG	0.952937
12	Tesla	PER	0.859428
13	G	ORG	0.428059
14	##iga	LOC	0.340912
15	##fa	ORG	0.637638
16	Berlin	LOC	0.999466
17	Germany	LOC	0.999675
18	Jeff Bezos	PER	0.981416

19	Amazon	ORG	0.998892
20	Andy Jassy	PER	0.999622
21	Seattle	LOC	0.998850
22	World Health Organization	ORG	0.998904
23	Geneva	LOC	0.997813
24	Switzerland	LOC	0.999636
25	C	PER	0.999553
26	##rist	PER	0.984562
27	##iano Ronaldo	PER	0.900957
28	Al Nassr	ORG	0.996720
29	Riyadh	LOC	0.851128
30	Saudi Arabia	LOC	0.999536
31	Google	ORG	0.998172
32	Bangalore	LOC	0.999238
33	India	LOC	0.999594
34	Sun	PER	0.999388
35	##dar Pichai	PER	0.919205
36	European Central Bank	ORG	0.999453
37	Frankfurt	LOC	0.998580
38	Germany	LOC	0.999677
39	Taylor Swift	PER	0.999329
40	Wembley Stadium	LOC	0.996156
41	London	LOC	0.999592

Run 2: biography-style paragraphs

paragraphs_run2 = [

"Marie Curie conducted groundbreaking research on radioactivity at the
↪University of Paris.",

"Nelson Mandela became the first Black president of South Africa after
↪decades of activism in Johannesburg.",

"Albert Einstein developed the theory of relativity while working at the
↪Swiss Patent Office in Bern.",

"Malala Yousafzai received the Nobel Peace Prize after her advocacy work in
↪Pakistan and at the United Nations.",

"Barack Obama served as President of the United States and later opened the
↪Obama Foundation in Chicago.",

"Mahatma Gandhi led the independence movement against British rule across
↪India.",

"Angela Merkel served as Chancellor of Germany and represented the country
↪at the G7 summit in Munich.",

"Jane Goodall founded the Jane Goodall Institute after her research on
↪chimpanzees in Tanzania.",

"Nikola Tesla worked for Westinghouse Electric before establishing his own
↪laboratory in New York.",

"Kofi Annan served as Secretary-General of the United Nations, based in New
↪York City.",

```
]
df_ner_run2 = run_ner_experiment("bert_ner_run2_bios", paragraphs_run2)
df_ner_run2
```

```
[bert_ner_run2_bios] total entities = 36 | distribution = {'PER': 13, 'LOC': 12,
'ORG': 7, 'MISC': 4}
```

	paragraph_id	paragraph \
0	0	Marie Curie conducted groundbreaking research ...
1	0	Marie Curie conducted groundbreaking research ...
2	1	Nelson Mandela became the first Black presiden...
3	1	Nelson Mandela became the first Black presiden...
4	1	Nelson Mandela became the first Black presiden...
5	1	Nelson Mandela became the first Black presiden...
6	2	Albert Einstein developed the theory of relati...
7	2	Albert Einstein developed the theory of relati...
8	2	Albert Einstein developed the theory of relati...
9	3	Malala Yousafzai received the Nobel Peace Priz...
10	3	Malala Yousafzai received the Nobel Peace Priz...
11	3	Malala Yousafzai received the Nobel Peace Priz...
12	3	Malala Yousafzai received the Nobel Peace Priz...
13	4	Barack Obama served as President of the United...
14	4	Barack Obama served as President of the United...
15	4	Barack Obama served as President of the United...
16	4	Barack Obama served as President of the United...
17	5	Mahatma Gandhi led the independence movement a...
18	5	Mahatma Gandhi led the independence movement a...
19	5	Mahatma Gandhi led the independence movement a...
20	5	Mahatma Gandhi led the independence movement a...
21	5	Mahatma Gandhi led the independence movement a...
22	6	Angela Merkel served as Chancellor of Germany ...
23	6	Angela Merkel served as Chancellor of Germany ...
24	6	Angela Merkel served as Chancellor of Germany ...
25	6	Angela Merkel served as Chancellor of Germany ...
26	7	Jane Goodall founded the Jane Goodall Institut...
27	7	Jane Goodall founded the Jane Goodall Institut...
28	7	Jane Goodall founded the Jane Goodall Institut...
29	8	Nikola Tesla worked for Westinghouse Electric ...
30	8	Nikola Tesla worked for Westinghouse Electric ...
31	8	Nikola Tesla worked for Westinghouse Electric ...
32	9	Kofi Annan served as Secretary-General of the ...
33	9	Kofi Annan served as Secretary-General of the ...
34	9	Kofi Annan served as Secretary-General of the ...
35	9	Kofi Annan served as Secretary-General of the ...

entity label	score
--------------	-------

0	Marie Curie	PER	0.989940
1	University of Paris	ORG	0.994023
2	Nelson Mandela	PER	0.999418
3	Black	MISC	0.998598
4	South Africa	LOC	0.999047
5	Johannesburg	LOC	0.999522
6	Albert Einstein	PER	0.998725
7	Swiss Patent Office	ORG	0.984982
8	Bern	LOC	0.995165
9	Malala Yousafzai	PER	0.974375
10	Nobel Peace Prize	MISC	0.998008
11	Pakistan	LOC	0.999823
12	United Nations	ORG	0.998713
13	Barack Obama	PER	0.999485
14	United States	LOC	0.999385
15	Obama Foundation	ORG	0.994400
16	Chicago	LOC	0.999197
17	Ma	PER	0.999624
18	##hat	PER	0.824744
19	##ma Gandhi	PER	0.979931
20	British	MISC	0.999629
21	India	LOC	0.999398
22	Angela Merkel	PER	0.997748
23	Germany	LOC	0.999655
24	G7	MISC	0.995469
25	Munich	LOC	0.999058
26	Jane Goodall	PER	0.999445
27	Jane Goodall Institute	ORG	0.990127
28	Tanzania	LOC	0.999741
29	Nikola Tesla	PER	0.998651
30	Westinghouse Electric	ORG	0.998191
31	New York	LOC	0.999381
32	Ko	PER	0.999626
33	##fi Annan	PER	0.984989
34	United Nations	ORG	0.999102
35	New York City	LOC	0.999522

Observations: The news batch (Run 1) yielded **42 entities** (LOC 19, PER 11, ORG 11, MISC 1) while the biography batch (Run 2) yielded **36 entities** (PER 13, LOC 12, ORG 7, MISC 4) — biographies mention more people and fewer organizations than news wire text, which is skewed toward companies/institutions. The model correctly tags most well-known names, but subword tokenization causes visible errors: uncommon/compound names get split into partial fragments ("Ma", "##hat", "##ma Gandhi"; "Ko", "##fi Annan"; "C", "##rist", "##iano Ronaldo"), and one clear mislabel occurs where "Elon Musk" is tagged ORG and "Tesla" is tagged PER (the two labels are swapped) with much lower confidence (0.95 and 0.86) than the typical >0.99 seen on correctly tagged entities — low confidence is a useful signal for flagging likely NER mistakes.

1.4 Task 3: Text Generation

Use the pretrained `gpt2` model to generate text continuations for five prompts. Two runs are logged to MLflow with different sampling parameters (`temperature`, `max_length`) so the effect of each parameter on the generated text can be compared; each run logs its parameters, the generated outputs as a text artifact, and a short observations file as a second artifact.

```
generator = pipeline("text-generation", model="gpt2", device=-1)

prompts = [
    "The future of artificial intelligence is",
    "In a small village by the mountains,",
    "The best way to learn a new skill is",
    "Scientists recently discovered that",
    "On a rainy afternoon, she decided to",
]

def run_generation_experiment(run_name, temperature, max_length, seed=42):
    set_seed(seed)
    outputs = []
    for prompt in prompts:
        result = generator(
            prompt,
            max_length=max_length,
            max_new_tokens=None, # avoid pipeline default overriding max_length
            temperature=temperature,
            do_sample=True,
            num_return_sequences=1,
            truncation=True,
            pad_token_id=generator.tokenizer.eos_token_id,
        )
        outputs.append(result[0]["generated_text"])

    output_text = "\n\n".join(
        f"Prompt: {p}\nGenerated: {o}" for p, o in zip(prompts, outputs)
    )
    artifact_path = f"{run_name}_generations.txt"
    with open(artifact_path, "w") as f:
        f.write(output_text)

    avg_len = sum(len(o) for o in outputs) / len(outputs)
    observations = (
        f"Run: {run_name}\n"
        f"temperature={temperature}, max_length={max_length}\n"
        f"Average generated length (characters): {avg_len:.1f}\n"
    )
    obs_path = f"{run_name}_observations.txt"
```

```

with open(obs_path, "w") as f:
    f.write(observations)

with mlflow.start_run(run_name=run_name):
    mlflow.log_param("model_name", "gpt2")
    mlflow.log_param("temperature", temperature)
    mlflow.log_param("max_length", max_length)
    mlflow.log_param("num_prompts", len(prompts))
    mlflow.log_metric("avg_output_length_chars", avg_len)
    mlflow.log_artifact(artifact_path)
    mlflow.log_artifact(obs_path)

os.remove(artifact_path)
os.remove(obs_path)

for p, o in zip(prompts, outputs):
    print(f"PROMPT: {p}\nGENERATED: {o}\n")
return outputs

mlflow.set_experiment("A09_Task3_Text_Generation")

```

Loading weights: 100% | 148/148 [00:00<00:00, 7405.92it/s]

```

<Experiment:
artifact_location=('/Users/vishaniraveendran/Library/CloudStorage/GoogleDrive-
vishani.20260276@iit.ac.lk/My '
'Drive/MS-C-Vish/First Year/Ai Tools and '
'Concepts/Assignment/Assignments/A09/mlruns/3'), creation_time=1785590158956,
effective_trace_archival_retention=None, experiment_id='3',
last_update_time=1785590158956, lifecycle_stage='active',
name='A09_Task3_Text_Generation', tags={}, trace_location=None,
workspace='default'>

```

```

# Run 1: low temperature, shorter output -> more focused/conservative text
outputs_run1 = run_generation_experiment("gpt2_run1_low_temp", temperature=0.7,
↳max_length=50)

```

PROMPT: The future of artificial intelligence is
GENERATED: The future of artificial intelligence is uncertain, but a new report
in the journal Proceedings of the National Academy of Sciences shows that we can
learn from our mistakes.

The study, led by neuroscientist Alan Gershman at Queen's University

PROMPT: In a small village by the mountains,
GENERATED: In a small village by the mountains, a family has gathered in their

summer camp to celebrate. At the end of the day, the locals are the only people in their community. But the village is not the only one. The people of the village

PROMPT: The best way to learn a new skill is

GENERATED: The best way to learn a new skill is to learn what you already know. That means getting a sense of what others are thinking about you and what they think you're doing.

You may have read articles about how to communicate your message,

PROMPT: Scientists recently discovered that

GENERATED: Scientists recently discovered that the same bacteria can grow in the open and on the sides of buildings, and that this process is linked to different types of bacteria.

The research was published in the journal Science.

"Our findings suggest that bacteria

PROMPT: On a rainy afternoon, she decided to

GENERATED: On a rainy afternoon, she decided to watch the sunset, feeling a sense of excitement at the spectacle. "I'd never seen it in my life before," she said. "It was very romantic."

But for Kailash, a

```
# Run 2: high temperature, longer output -> more random/creative text
outputs_run2 = run_generation_experiment("gpt2_run2_high_temp", temperature=1.
↪3, max_length=100)
```

PROMPT: The future of artificial intelligence is

GENERATED: The future of artificial intelligence is uncertain and could ultimately create massive technological gaps."

PROMPT: In a small village by the mountains,

GENERATED: In a small village by the mountains, they are already moving toward a new city...

There was a great deal of discontent in the capital and its rural population. The town had been ravaged when Hohin, a leader in his father's administration after the Battle of Endless Sea, was killed with his party by a strong party loyal to Hohin. The town in which Hohin lived belonged to the nobility who became its new rulers, while the town had belonged to others,

PROMPT: The best way to learn a new skill is

GENERATED: The best way to learn a new skill is to learn it while trying to stay

focused. Learn it, if you like, as many times as you possibly can during school."

The goal for learning new skills is to improve your overall performance, which typically isn't very often for a white kid when studying: students look to study more to get to the upper-ranked classes.

According to Nix, when black kids are struggling at their secondary schools, it's also the school in

PROMPT: Scientists recently discovered that

GENERATED: Scientists recently discovered that the human immune (in our cells) fights harmful molecules in water. Some people believe we're not immune to the sun, as humans believe our bodies produce a lot of chlorine, but if you see blue water in the air a little over a million years from us then that chemical isn't present in the sun. It only happens when our eyes work just right after you turn sunlight away from us. "People like sun-deprived food sources, sun-degrading chemical

PROMPT: On a rainy afternoon, she decided to

GENERATED: On a rainy afternoon, she decided to leave her seat along by the edge of a grass edge at 4 o'clock a.m. She called on that one with those few remaining miles.

She pulled inside by 10:28 a.m."I couldn't believe the pain, like the pain has gone," Ecker said.

Ecker was looking pretty good for a long time before the ambulance drove onto the property and she died. No one else could believe it! Just

Observations: Run 1 (temperature=0.7, maxlength=50) produces short, coherent, on-topic continuations that stay grounded in the prompt (average ~230 characters). Run 2 (temperature=1.3, max_length=100) produces noticeably longer (~380 characters) and more diverse/creative text, but coherence degrades — it drifts topic more freely, invents unrelated details, and occasionally produces disjointed or nonsensical passages. This confirms the expected trade-off: lower temperature favors focused, predictable text, while higher temperature favors variety and creativity at the cost of factual/logical consistency. (Note: the pipeline call explicitly passes max_new_tokens=None alongside max_length — without this, newer versions of transformers silently let a default max_new_tokens override max_length, which caused the two runs to ignore the intended length setting.)__

1.5 Viewing the MLflow Experiments

All six runs (two per task) are stored in ./mlflow.db (SQLite) in this folder. To inspect them in the MLflow UI, run the following from the A09 directory and open the printed URL in a browser:

```
mlflow ui --backend-store-uri sqlite:///mlflow.db
```

Each experiment (A09_Task1_Sentiment_Classification, A09_Task2_Named_Entity_Recognition,

A09_Task3_Text_Generation) contains two runs that can be compared side by side using the parameters and metrics logged above.